Program 3

Code a Dockerised Python Flask or Node.js Application

Project Structure
> Dockerfile
> requirements.txt
> app.py

Step 1 : Create a Docker file

```
 GNU nano 7.2
#Author : Uday R
#created on :


# Using Python base image
FROM python:3.9-slim

# Setting working directory
WORKDIR /app

# Copying the requirements file and install dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copying the application code
COPY . .

# Setting Flask environment variables
ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

# Exposing the port
EXPOSE 5000

# Command to run flask
CMD ["flask", "run"]
```

Step 2 : Create Python file app.py

```
 GNU nano 7.2
from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello, Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Step 3 : Create a Sample text file requirements.txt and add the following

flask

Step 4 : Execute Docker Build command

```
1rv24mc111_uday@udayraj-HP-Laptop-15s-eq2xxx:~/lab/prog3$ sudo nano Dockerfile
1rv24mc111_uday@udayraj-HP-Laptop-15s-eq2xxx:~/lab/prog3$ docker build -t prog3 .
[+] Building 8.6s (11/11) FINISHED                                          docker:default
 => [internal] load build definition from Dockerfile                                  0.0s
 => => transferring dockerfile: 454B                                                  0.0s
 => [internal] load metadata for docker.io/library/python:latest                      1.5s
 => [internal] load .dockerignore                                                     0.0s
 => => transferring context: 2B                                                       0.0s
 => [1/6] FROM docker.io/library/python:latest@sha256:92b51c6fa86611bbf2000abf360a206837cee8071b382119990eb48ee3953827  0.0s
 => [internal] load build context                                                     0.0s
 => => transferring context: 514B                                                     0.0s
 => CACHED [2/6] WORKDIR /app                                                         0.0s
 => CACHED [3/6] COPY requirments.txt .                                               0.0s
 => CACHED [4/6] RUN pip install --upgrade pip                                        0.0s
 => [5/6] RUN pip install --no-cache-dir -r requirments.txt                           6.9s
 => [6/6] COPY . .                                                                    0.0s
 => exporting to image                                                                0.2s
 => => exporting layers                                                               0.2s
 => => writing image sha256:3ef0ae791d95b0bc3c578827981c2dfe4d6c4069d9f1c7e064cca542a2250ac3  0.0s
 => => naming to docker.io/library/prog3                                              0.0s
1rv24mc111_uday@udayraj-HP-Laptop-15s-eq2xxx:~/lab/prog3$
```

Step 5 : Run the docker run command

```
1rv24mc111_uday@udayraj-HP-Laptop-15s-eq2xxx:~/lab/prog1$ docker run -d -p 5000:5000 --name cont prog1
e4ad74f55e780b0cc46808ef8ef36857e7bac165c1f56317b845264a7453f4e3
```

Step 6 : In Web Browser open http://localhost:5000